

Tyler Gorda

ASSISTANT PROFESSOR · THE OHIO STATE UNIVERSITY

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Research Interests

Dense nuclear matter Bulk and transport properties of dense quark matter

Neutron stars Equation of state of neutron-star matter, physics of neutron-star mergers

In-medium field theory Equation of state of dense/hot Quantum Chromodynamic matter, transport and energy loss in a dense/hot medium

Education

Doctor of Philosophy in Physics

Boulder, Colorado, USA

UNIVERSITY OF COLORADO BOULDER

Aug. 2011-Dec. 2016

- Supervisor: Prof. Paul Romatschke
- Thesis: "From pQCD to neutron stars: matching equations of state to constrain global star properties" (arXiv:1608.04358)

Master of Science in Physics

Boulder, Colorado, USA

UNIVERSITY OF COLORADO BOULDER

Aug. 2011- Dec. 2014

- 4.000 GPA

Bachelor of Science in Physics

New Brunswick, New Jersey, USA

RUTGERS, THE STATE UNIVERSITY OF NEW JERSEY

Aug. 2007-May 2011

- Summa cum laude, 3.976 GPA
- Double major in Physics and Mathematics

Work Experience

Assistant Professor

Columbus, Ohio, USA

THE OHIO STATE UNIVERSITY

Aug. 2025 - PRESENT

Postdoctoral Researcher

Frankfurt, Germany

GOETHE UNIVERSITY FRANKFURT, INSTITUTE FOR THEORETICAL PHYSICS

Nov. 2023 - July 2025

- Topics: QCD at high density, Physics of neutron-star mergers, Astrophysics of high-density QCD, Thermal field theory

Postdoctoral Researcher

Darmstadt, Germany

TU DARMSTADT, INSTITUTE FOR NUCLEAR PHYSICS

Sept. 2020 - Sept. 2023

- Topics: QCD at high density, Astrophysics of high-density QCD, Thermal field theory

Postdoctoral Researcher

Charlottesville, Virginia, USA

UNIVERSITY OF VIRGINIA

Aug. 2018 - Sept. 2020

- Topics: Thermal field theory, QCD at high density, Astrophysics of high-density QCD, Particle Propagation in a QCD medium

Postdoctoral Researcher

Helsinki, Finland

UNIVERSITY OF HELSINKI

Aug. 2016 - Aug. 2018

- Topics: Thermal field theory, QCD at high density, Astrophysics of high-density QCD, Particle cosmology

Research assistant

UNIVERSITY OF COLORADO BOULDER

Boulder, Colorado, USA

Oct. 2012-May 2016

- Topics: Thermal field theory, QCD at high density, Astrophysics of high-density QCD, AdS-CFT correspondence, collective flow in heavy-ion collisions

Funding and Grants

Gauss Centre for Supercomputing

PROJECT EXTENSION OF LARGE SCALE PROJECT "MICROPHYSICAL ASPECTS OF BINARY NEUTRON STAR MERGERS", PI L. REZZOLLA.

100 MILLION CPU HOURS AWARDED ON THE HAWK SUPERCOMPUTER.

Stuttgart, Germany

Jan. 2025 - Jan. 2026

Mentoring & Supervision

Alexandra Semposki Postdoctoral Researcher.

THE OHIO STATE UNIVERSITY

Columbus, Ohio

May 2026 - present

El Hajj Malick Diagne Junior High School Student.

METRO EARLY COLLEGE HIGH SCHOOL

Columbus, Ohio

Feb. 2026 - May 2026

- Completed a *Capstone Project* on the Neutron-star equation of state. Developed an open-source TOV solver + Web interface (**EoS-Explorer**), based on python and streamlit

Swastick Majumder PhD student.

THE OHIO STATE UNIVERSITY

Columbus, Ohio

Sept. 2025 - present

Sofia Blomqvist MSc and PhD student. Official co-advisor for MSc

UNIVERSITY OF HELSINKI

Helsinki, Finland

June 2023 - Aug. 2025

Andreas Geißel PhD student. Advised and mentored

TU DARMSTADT

Darmstadt, Germany

April 2023 - July 2025

Oleg Komoltsev PhD student. Advised and mentored

UNIVERSITY OF STAVANGER

Stavanger, Norway

Jan. 2022 - May 2025

Saga Säppi MSc and PhD student. Advised and mentored

UNIVERSITY OF HELSINKI

Helsinki, Finland

Aug. 2016 - July. 2018

Eemeli Annala MSc and PhD student. Advised and mentored

UNIVERSITY OF HELSINKI

Helsinki, Finland

Aug. 2016 - July. 2018

Teaching Experience

Teaching Assistant

THE OHIO STATE UNIVERSITY

Columbus, Ohio, USA

Sept. 2025-Dec. 2025

- Undergraduate introductory mechanics for engineers (Physics 1250)

Teaching Assistant

GOETHE UNIVERSITÄT FRANKFURT AM MAIN

Frankfurt, Germany

Oct. 2024-Feb. 2025

- General Relativity

Lectures on perturbative QCD at high temperatures and density

QCD MASTER CLASS 2023

Saint-Jacut-de-la-Mer, France

June 2023

- **10 hours of lectures** to advanced PhD students, postdocs, and permanent researchers on the formalism of perturbative QCD in medium
- Topics covered: (i) framework of relativistic thermal and high-density perturbation theory, (ii) infrared problems in thermal field theory and their resolution for the pressure, (iii) the general structure of the perturbative QCD pressure, and (iv) current status of theoretical calculations.

Lectures on Neutron Stars and the Equation of State of Dense Matter

Graz, Austria

DOKTORATSKOLLEG PARTICLES AND INTERACTIONS PH.D. RETREAT

May 2022

- **3 hours of lectures**, as part of a three-day retreat for doctoral students
- Topics covered: (i) General properties of neutron stars and their observation (ii) theoretical techniques for computing the thermodynamic properties of dense matter (iii) current status of astrophysical and theoretical constraints on the behavior of dense matter.

Lectures on Perturbative QCD at high densities

Santiago de Compostela, Spain

INSTITUTO GALEGO DE FISICA DE ALTAS ENERGIAS, UNIVERSITY OF SANTIAGO DE COMPOSTELA (ONLINE)

Nov. 2021

- **2 hours of lectures**, as part of course on Neutron-Star physics
- Topics covered: (i) framework of relativistic thermal and high-density perturbation theory, (ii) infrared problems in thermal field theory and their resolution for the pressure, (iii) the general structure of the perturbative QCD pressure, and (iv) current status of theoretical calculations.

Teaching Assistant

Boulder, Colorado, USA

UNIVERSITY OF COLORADO BOULDER

2011, 2014-2016

- Undergraduate introductory mechanics for majors
- Undergraduate introductory electromagnetism for majors
- Undergraduate introductory mechanics for non-majors
- Undergraduate introductory electromagnetism for non-majors

Presentations & Talks

In total: 3 colloquia, 16 plenary talks, 26 invited talks, 16 seminars, 8 contributed talks.

Recent or notable talks include:

NT/RBRC Seminar, Brookhaven National Lab

Brookhaven, New York, USA

INVITED SEMINAR TALK “Phase quenched QCD: A new approach to determine the thermodynamics of deconfined matter to high accuracy”

March 2026

NT Seminar, University of Maryland

College Park, Maryland, USA

INVITED SEMINAR TALK “Phase quenched QCD: A new approach to determine the thermodynamics of deconfined matter to high accuracy”

March 2026

Network for Neutrinos, Nuclear Astrophysics, and Symmetries (N3AS) online seminar

(Virtual Event)

INVITED SEMINAR (ONLINE) TALK “Constrained Brownian-bridge prior for neutron-star equation-of-state inference”

December 2025

Nuclear Physics from Multi-Messenger Mergers (N3PM) online seminar

(Virtual Event)

INVITED SEMINAR (ONLINE) TALK “Constrained Brownian-bridge prior for neutron-star equation-of-state inference”

December 2025

The Ohio State University

Columbus, Ohio, USA

INVITED SEMINAR TALK “Constrained Brownian-bridge prior for neutron-star equation-of-state inference”

December 2025

The Ohio State University

Columbus, Ohio, USA

INVITED COLLOQUIUM “Probing strongly interacting matter at the highest densities”

February 2025

Institute of Astronomy and Astrophysics, Université libre de Bruxelles

Brussels, Belgium

INVITED SEMINAR (ONLINE) “Constraining the behavior of strongly interacting matter at the highest densities”

February 2025

The 24th Zimányi school winter workshop on heavy ion physics

Budapest, Hungary

INVITED PLENARY TALK “Quark matter and Nuclear astrophysics: Recent developments”

December 2024

The 3rd APCTP-Triumf Joint Workshop: From Nuclei to Neutron Stars

Busan, South Korea

INVITED PLENARY TALK “Constraining strongly interacting matter at the highest densities with perturbative QCD”

September 2024

New aspects of nuclear physics and nuclear astrophysics

Seoul, South Korea

INVITED PLENARY TALK “Quark matter in the cores of massive neutron stars”

September 2024

INT-24-89W: EOS Measurements with Next-Generation Gravitational-Wave Detectors

Seattle, Washington, USA

PLENARY TALK “Listening to the long ringdown”

September 2024

Strong and Electro-weak Matter 2024

Frankfurt, Germany

PLENARY TALK “Pressure and speed of sound in (two-flavor) color-superconducting quark matter at NLO”

August 2024

CRC-TR 211 (Strong-interaction matter under extreme conditions) Meeting

Frankfurt, Germany

INVITED PLENARY TALK “Listening to the long ringdown”

July 2024

Heidelberg University

Heidelberg, Germany

INVITED THEORY COLLOQUIUM “Constraining the behavior of strongly interacting matter at the highest densities”

June 2024

ELEMENTS Annual Conference 2024

Frankfurt, Germany

INVITED COLLOQUIUM “Constraining the behavior of strongly interacting matter at the highest densities”

April 2024

Institute for Nuclear Theory

Seattle, Washington, USA

INVITED S@INT SEMINAR “Probing the behavior of strongly interacting matter at the highest densities”

April 2024

University of Michigan

Ann Arbor, Michigan

SEMINAR “Constraining the microphysics of high-density QCD”

March 2024

Frankfurt Institute for Advanced Studies

Frankfurt, Germany

SEMINAR “Constraining the microphysics of high-density QCD”

October 2023

Quark Matter 2023

Houston, Texas, USA

INVITED PLENARY TALK, “Quark Matter and Nuclear Astrophysics”

September 2023

ELEMENTS Annual Conference 2023

Bad Nauheim, Germany

PLENARY TALK, “Constraints on strong phase transitions in neutron stars”

July 2023

University of Southampton

Southampton, UK

SEMINAR “Constraining the microphysics of high-density QCD”

May 2023

Rencontres de Moriond: Gravitation

La Thuile, Italy

INVITED PLENARY TALK, “What multimessenger observations have told us about the EoS of NS matter”

Mar. 2023

Strong and Electro-weak Matter 2022

Saclay, France

INVITED PLENARY TALK, “Bayesian constraints on the neutron-star equation of state with QCD input”

June 2022

APS April Meeting (Session Q04: Mergers of Neutron Stars: Nuclear Physics from Gravitational Waves)

New York City, New York USA

INVITED TALK, “High-Density Quark Matter in the Cores of Neutron Stars”

Apr. 2022

Service

Physics Community Committee member

Aug. 2025-PRESENT

THE OHIO STATE UNIVERSITY

Astrocoffee Seminar Series Organizer

Jan. 2024-July 2025

SEMINAR SERIES AT GOETHE UNIVERSITY ON RECENT DEVELOPMENTS IN ASTROPHYSICS, COSMOLOGY, AND ADJACENT TOPICS

Member of the IReNA Online Seminar Organizing Committee

INTERNATIONAL RESEARCH NETWORK FOR NUCLEAR ASTROPHYSICS

Sept. 2021-May 2022

Refereed Publications

VARIOUS JOURNALS

2018-PRESENT

- Physical Review Letters
- Physical Review D
- Physics Letters B
- Monthly Notices of the Royal Astronomical Society Letters
- Journal of High Energy Physics (JHEP)
- International Journal of Modern Physics A
- Universe
- European Physical Journal C

Founding member of the Graduate Liaison Committee

FOUNDED TO BETTER SUPPORT GRADUATE STUDENTS IN THEIR COMMUNICATION WITH THE PHYSICS DEPARTMENT

University of Colorado Boulder, USA

2014-2016

Coordinated Programs and Professional Societies

Member of the Collaborative Research Center TransRegio 211 (Strong-interaction matter under extreme conditions)

April 2022-Aug. 2025

COORDINATED PROGRAM BETWEEN BIELEFELD UNIVERSITY, TU DARMSTADT, AND GOETHE UNIVERSITY, TO DECISIVELY ADVANCE OUR UNDERSTANDING OF STRONG-INTERACTION MATTER UNDER EXTREME CONDITIONS OF TEMPERATURE AND DENSITY

Member of ELEMENTS: Exploring the Universe from Microscopic to Macroscopic Scales

April 2022-March 2025

CLUSTER PROJECT BETWEEN GOETHE UNIVERSITY, TU DARMSTADT, JLU GIESSEN, AND GSI/FAIR TO ADDRESS THE QUESTION OF THE ORIGIN OF THE HEAVY CHEMICAL ELEMENTS, SUCH AS PLATINUM AND GOLD, IN OUR UNIVERSE.

Member of the American Physical Society

Feb. 2022-PRESENT

Outreach & Media Coverage

Astronews.com article (in German) about my work on Pairing effects on equation of state at high density quark matter *Phys. Rev. Lett.* 135.21 (2025), 211901

Jan. 2026

Phys.org article about my work on the Long ringdown signal in binary neutron-star mergers *Nature Commun.* 16.1 (2025), 1320

Feb. 2025

Helsingin Sanomat (largest newspaper in Finland), Phys.org, Universe Today, Space Daily articles about my work *Phys. Rev. Lett.* 133.7 (2024), 071901

Aug. 2024

Phys.org (2 articles), Yahoo News, Space.com, Universe Today, Sci.News articles about my work *Nat Commun.* 2023

Dec./Jan. 2023

PNAS article mentioning my work *Nature Phys.* 2020

Nov. 2020

Podcast Interview with *The Cosmic Companion*

June 2020

DISCUSSING EVIDENCE FOR QUARK-MATTER CORES IN MASSIVE NEUTRON STARS IN *Nature Phys.* 2020

Phys.org, Physicsworld, Medium, and IFLS articles

June 2020

MORE ARTICLES ABOUT MY WORK *Nature Phys.* 2020

Phys.org article about my work on the Two-Higgs-Doublet Model *Phys. Rev. Lett.* **121** 191802 (2018)

Nov. 2018

Sky and Telescope article mentioning my work constraining the neutron-star-matter equation of state using gravitational waves *Phys. Rev. Lett.* **120** 172702 (2018)

May 2018

APS Synopsis article on the same work *Phys. Rev. Lett.* **120** 172702 (2018)

Apr. 2018

Software/Programming Experience

Programming Extensive experience with Mathematica and **Python**, and good working knowledge of **C/C++**

General IT $\text{\LaTeX}$ (typesetting), GNU/Linux (operating system), Git (version control)

Languages

English Mother tongue

German Level A2.2

French Approximate Level B1

Publications (since 01/2023) — Tyler Gorda

Summary of papers on [Inspire-HEP](#). As of July 2026: 3,647, citations, h-index 24

Note that **alphabetical ordering** of authors is the community standard in theoretical high-energy physics

Peer-Reviewed Articles

- [13] S. Blomqvist, C. Ecker, **T. Gorda**, and A. Vuorinen. “Strong model-agnostic constraints for twin-star solutions”. *Phys. Rev. D* 113.11 (2026), L111504. arXiv: 2512.19477 [astro-ph.HE].
- [12] **T. Gorda**, O. Komoltsev, A. Kurkela, and E. Sunde. “Constrained Gaussian-process-bridge Prior for Neutron-star Equation-of-state Inference”. *Astrophys. J.* 1002.1 (2026), 40. arXiv: 2512.18044 [astro-ph.HE].
- [11] A. Geißel, **T. Gorda**, and J. Braun. “Color Superconductivity under Neutron-Star Conditions at Next-to-Leading Order”. *Phys. Rev. Lett.* 135.21 (2025), 211901. arXiv: 2504.03834 [hep-ph].
- [10] A. Geißel, **T. Gorda**, and J. Braun. “Pressure and speed of sound in two-flavor color-superconducting quark matter at next-to-leading order”. *Phys. Rev. D* 110.1 (2024), 014034. arXiv: 2403.18010 [hep-ph].
- [9] C. Ecker, **T. Gorda**, A. Kurkela, and L. Rezzolla. “Constraining the equation of state in neutron-star cores via the long-ringdown signal”. *Nature Commun.* 16.1 (2025), 1320. arXiv: 2403.03246 [astro-ph.HE].
- [8] J. Cruz Rojas, **T. Gorda**, C. Hoyos, N. Jokela, M. Järvinen, A. Kurkela, R. Paatelainen, S. Säppi, and A. Vuorinen. “Estimate for the Bulk Viscosity of Strongly Coupled Quark Matter Using Perturbative QCD and Holography”. *Phys. Rev. Lett.* 133.7 (2024), 071901. arXiv: 2402.00621 [hep-ph].
- [7] O. Komoltsev, R. Somasundaram, **T. Gorda**, A. Kurkela, J. Margueron, and I. Tews. “Equation of state at neutron-star densities and beyond from perturbative QCD”. *Phys. Rev. D* 109.9 (2024), 094030. arXiv: 2312.14127 [nucl-th].
- [6] G. D. Moore and **T. Gorda**. “Bounding the QCD Equation of State with the Lattice”. *JHEP* 12 (2023), 133. arXiv: 2309.15149 [nucl-th].
- [5] **T. Gorda**, R. Paatelainen, S. Säppi, and K. Seppänen. “Equation of State of Cold Quark Matter to $O(\alpha_s^3 \ln \alpha_s)$ ”. *Phys. Rev. Lett.* 131.18 (2023), 181902. arXiv: 2307.08734 [hep-ph].
- [4] **T. Gorda**, R. Paatelainen, S. Säppi, and K. Seppänen. “Soft gluon self-energy at finite temperature and density: hard NLO corrections in general covariant gauge”. *JHEP* 08 (2023), 021. arXiv: 2304.09187 [hep-ph].
- [3] E. Annala, **T. Gorda**, J. Hirvonen, O. Komoltsev, A. Kurkela, J. Nättilä, and A. Vuorinen. “Strongly interacting matter exhibits deconfined behavior in massive neutron stars”. *Nat Commun.* 14 (2023), 8451. arXiv: 2303.11356 [astro-ph.HE].
- [2] **T. Gorda**, O. Komoltsev, A. Kurkela, and A. Mazeliauskas. “Bayesian uncertainty quantification of perturbative QCD input to the neutron-star equation of state”. *JHEP* 06 (2023), 002. arXiv: 2303.02175 [hep-ph].
- [1] **T. Gorda**, K. Hebeler, A. Kurkela, A. Schwenk, and A. Vuorinen. “Constraints on Strong Phase Transitions in Neutron Stars”. *Astrophys. J.* 955.2 (2023), 100. arXiv: 2212.10576 [astro-ph.HE].

Preprints

- [3] **T. Gorda**, O. Komoltsev, A. Kurkela, and J. Schaffner-Bielich. “As above, so below: assessing extremeness of the neutron-star equation of state based on the unstable branch”. June 2026. arXiv: 2606.23929 [nucl-th].
- [2] O. Ferreira, E. S. Fraga, **T. Gorda**, R. Paatelainen, L. Sandbote, and K. Seppänen. “Static linear response of hot and dense QCD matter to electromagnetic fields: Leading hard and soft QCD corrections”. June 2026. arXiv: 2606.16377 [hep-ph].
- [1] **T. Gorda**, P. Navarrete, R. Paatelainen, L. Sandbote, and K. Seppänen. “A new approach to determine the thermodynamics of deconfined matter to high accuracy”. Nov. 2025. arXiv: 2511.09627 [hep-ph].

Proceedings

- [2] C. Ecker, **T. Gorda**, A. Kurkela, and L. Rezzolla. “Listening to the long ringdown: A novel way to pinpoint the EOS in neutron-star cores”. *31st International Conference on Ultra-relativistic Nucleus-Nucleus Collisions*. Sept. 2025. arXiv: 2509.18665 [astro-ph.HE].
- [1] **T. Gorda**. “Quark matter and nuclear astrophysics: recent developments”. *EPJ Web Conf.* 296 (2024), 01010. arXiv: 2312.09967 [nucl-th].